Beyond Entry: Employment Exits and Female Participation in India

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November 12, 2026

Abstract

Using rich panel micro-data on labor force participation status, we show that India's low female labor force participation reflects not only low entry into employment, but also extremely high exits. Employed women are seven times as likely as men to exit into non-participation, even conditional on demographics, including marriage and education. These exits are largely to home production and are persistent. We document that employment protections are associated with substantially lower exits for women—assigning exit rates of protected women to unprotected women could raise steady-state participation by 21% of the baseline.

Keywords: participation, transitions, female employment
JEL Codes: E24, J16, J21, J82

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1 Introduction

India's female labor force participation rate is among the lowest in the world, even conditional on its income rank. In this paper, we study the origins of this low participation rate by studying labor market flows. Figure 1 shows that this low level of female participation is explained both by anomalously lower entry and re-entry rates into employment from non-employment, i.e., the entry margin, and by a higher relative propensity of women to leave employment, i.e., the exit margin. On both entry and exit margins, the labor market dynamics of Indian men are similar to the median transition rates across countries, whereas women are outliers.

Evidence from developing countries have emphasized the role of domestic duties, restricted mobility, and social norms in determining women's entry to the labor force (Jayachandran, 2021). While much of the literature focuses on barriers to employment, we examine the dynamics of employment retention and show that exits into nonparticipation are an important component of urban India's gender participation gap. We show that this exit margin survives even after controlling for a range of observables, and that marital status, while important, cannot explain all of the participation gap. We also find that women who leave employment do so more persistently, and engage disproportionately in home production or childcare relative to men making similar employment transitions.

What then explains the higher exit rate of women relative to men? We argue that an important and underappreciated role is played by the quality of employment opportunities available, which we measure using the number of employment protections available on the job. Specifically, our data allows us to measure whether jobs (i) provide formal contracts, (ii) provide paid family leave, or (iii) provide social security contributions from the employer's side. We show that employment protection reduces exit rates for both men and women, but that they appear to matter substantially more for women. Moving from a job with one protection to all three would reduce the odds of a female exit to non-employment by nearly 50 percent, whereas the effect on male exits to non-employment is not statistically significant. This effect is not driven by the gender composition of jobs themselves, i.e., relative to men, women do not work in jobs which are less likely to be protected. Instead, we find that the exit rate for women is much more responsive to the absence of protections than that for men.

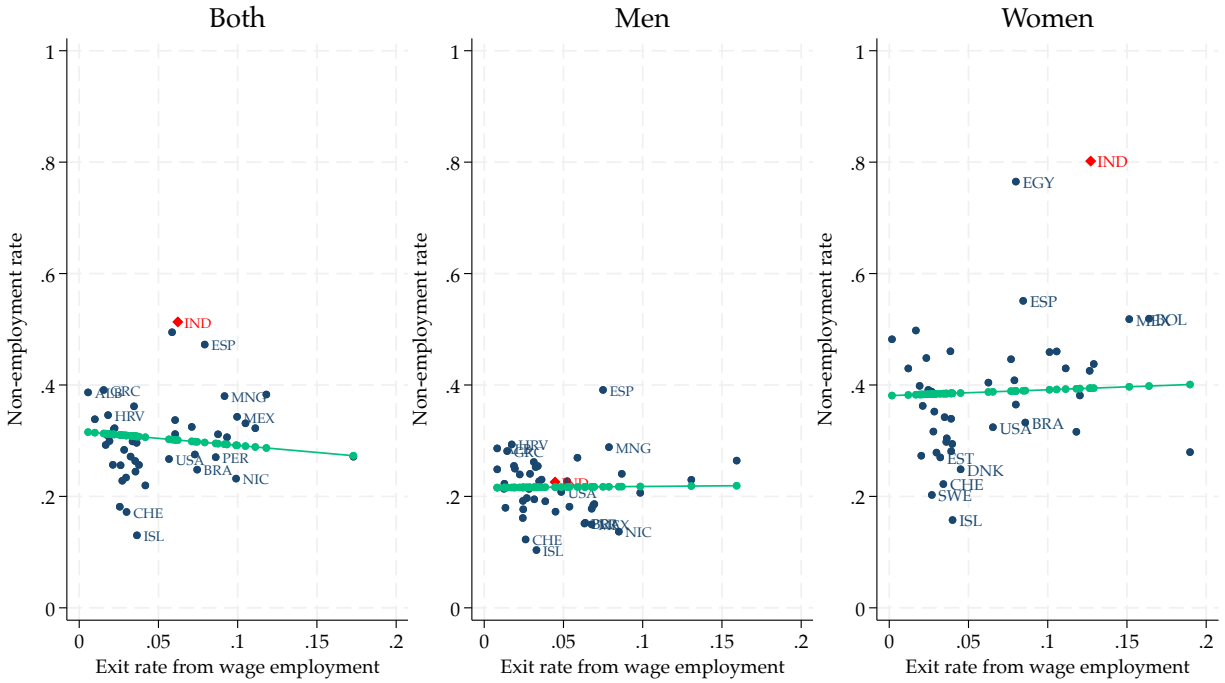
The final part of our analysis studies how important is the exit margin for female employment rates, and how much of female labor force participation and unemployment can be explained by employment protections and amenities. While much of the policy discussion focuses on addressing barriers to entry for women, our findings suggest that

retaining women who are already in the labor force is an equally important margin. In a series of steady-state counterfactuals, we find that equalizing women's employment exit rates to those of men raises female participation from 19 percent to 31 percent, an increase of over 50 percent. Importantly, substantial gains are possible even with providing more employment protection. Assigning women in unprotected jobs the exit rates of women with at least one employment protection would raise participation rates to 23 percent, a 21 percent increase, accounting for 40 percent of the gap in participation between men and women. We thus conclude that improving employment protections and job quality could be an important way for policymakers to increase female labor force participation in India.

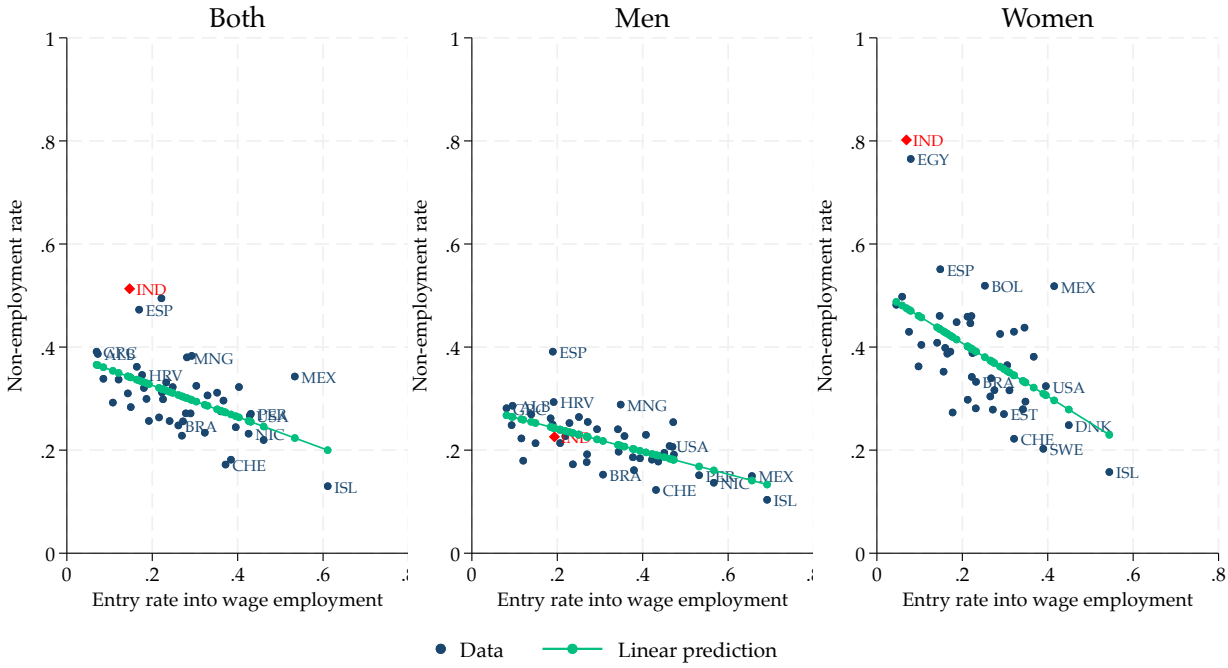
Related Literature Our paper contributes to three strands of literature. First, we contribute to the literature on how labor market flows shape aggregate participation and gender differences in labor force attachment. Elsby et al. (2019) show how entry and exit account for movements in labor force participation, while Albanesi and Şahin (2018) link changes in gender differences in unemployment to changes in labor force attachment. In developing economies, Donovan et al. (2023) find that workers enter and leave employment frequently, often moving between marginal jobs without securing more stable, better-paying work. We apply these perspectives to explaining India's gender participation gap, showing how frequent exits from employment and limited rates of re-entry contribute to explaining it.

We also contribute to a now voluminous literature seeking to explain India's low female participation. Klasen and Pieters (2015) examine household characteristics and labor demand, while Deshpande and Kabeer (2024) and Afridi et al. (2024) study domestic responsibilities, home production, and social norms. Closer to our work, Sarkar et al. (2019) examine women's employment entry and exit using the India Human Development Survey, and Deshpande and Singh (2021) document repeated labor force exits and reentries using CMIE panel data. We extend this work by linking quarterly employment transitions to job protections and quantifying their implications for aggregate participation. Distinguishing wage employment, self-employment, unemployment, and nonparticipation allows us to trace where women go after leaving work and how limited reentry contributes to low participation.

Finally, our findings on protections connect to research on gender differences in the value of job amenities. Wiswall and Zafar (2018) find that women value job stability and work flexibility more than men, and Faberman et al. (2026) study how differences in amenity valuations shape wages and employment. In India, Chakraborty and Mahajan



(a) Exit Margin



(b) Entry Margin

Figure 1: Job transition rates using data compiled by Donovan et al. (2023).

Notes: Pooled averages from 2010-2023 of non-employment rates plotted against exit rates from employment (panel A) and entry rates into employment (panel B), for the overall population (left), men (middle) and women (right). The non-employment rate is the share of people outside of labor force at the beginning of sample period for each country. All rates are computed as shares of individuals making that transition from date $t-1$ to date t as a share of individuals in the initial state at date $t-1$. Green lines plot predicted values conditioning on average GDP/capita. All data from Donovan et al. (2023).

(2026) document higher female employment shares in larger firms and discuss amenities as a possible explanation. Leveraging the fact that our data provides information both on employment dynamics and protection, we provide evidence that protections are associated with lower exit rates, particularly among women.

The rest of this paper proceeds as follows. In section 2, we describe the structure of our dataset. Section 3 documents labor market flow rates by gender and by type of protections available. Section 4 discusses the labor market destination of men and women who exit the labor force. Section 5 provides evidence that employment protection drastically reduces female exit rates to non-employment while leaving male exit rates virtually unchanged. Section 6 conducts counterfactuals replacing female exit rates by those for protected women, protected men and men in general. Section 7 concludes.

2 Data

Our analysis for India is based on the Periodic Labour Force Survey (PLFS) conducted by the National Statistical Office. The PLFS is a quarterly[1] survey of households with rotating panels, conducted since April 2017. It records extensive labor market information on all individuals residing within a household along with demographic and geographical identifiers. Each sampled household is interviewed in four consecutive quarters before rotating out of the panel, i.e., each quarter one fourth of existing households are replaced by a new set of households. This design allows us to track individuals across four consecutive quarters and thus collect information on up to three transitions across employment, unemployment and non-participation. While the survey includes rural households, these households are not re-surveyed in consecutive quarters and are thus, excluded from our sample.

Our sample period covers 2017:Q4 to 2024:Q2. We keep urban person-quarter pairs with valid Current Weekly Status (CWS) activity codes in both quarters, yielding about 3.6 million observations with approximately 134,000 observations in each quarter. We restrict our analysis only to prime-age individuals (18–59) to avoid capturing employment transition margins accounted by schooling and retirement. All estimates use the survey multiplier from the initial quarter as the transition weight.

Current labor market participation status is identified using the *current weekly status* (CWS) which records an individual's labor market activities in the reference week, where the reference week defined as the seven day period preceding the survey date. An individ-

[1] Since 2025, the PLFS has been conducted monthly; we end our analysis prior to this to avoid the time series break this generates.

ual is considered to be employed if they engage in at least one hour of work on any day within the reference period, or if they had a work opportunity for at least one hour on one day within the reference period but did not do the work due to illness or disability. A worker is considered unemployed if they did not engage in any economic activities, but either reported search effort or was available for work for at least one hour within the reference period. Non-employed workers are those who were not available for work at any point in the reference period.

We group CWS activity codes into three mutually exclusive states: employed (E), unemployed (U) and out of labor force (N). The "employed" state includes only regular and casual wage employees, excluding self-employed workers and unpaid family workers (codes: 31, 41, 42, 51, 71, 72, 98). For the "unemployed" state, we use the definition suggested by PLFS which includes CWS activity status 81, 82. All other activity codes are included within the activity status of out of labor force (N).

Besides labor market participation, PLFS also includes all relevant household characteristics including household size, relation to head of household, and residence state, and district, along with demographic characteristics of individuals residing with the household, including, but not limited to, gender, age, education, marital status, usual principal status, and daily activity status. Individuals reporting current status as employed (defined above) or self-employed are asked additional questions on job characteristics such as type of employment dwelling, job protections available, establishment size, industry, and occupation. Individuals who report being unemployed or out of labor force are asked questions about their efforts for job search and reason for leaving employment. We use information on job characteristics and reasons for unemployment to understand the mechanisms behind the large gender gap observed in labor force transition in India.

3 Gender Bias in Exits from Employment and Reentry

We first compare women's and men's movements into and out of wage employment. Table 1 reports the quarterly transition rates, subsequent outcomes among workers who leave employment, and differences in exit destinations after controlling for worker and job characteristics. The result is striking. Women who leave employment are far more likely than men to drop out of the labor force entirely. Nearly 5.2 percent of employed women separate from their employment and exit the labor force within a quarter while only 1.96 percent employed women exit to unemployment. In comparison, only 0.7 percent of employed men exit the labor force while 2.8 percent exit to unemployment within one quarter. Additionally, both unemployed women and women who are out of labor force are

significantly less likely to transition into employment relative to men within one quarter.[2]

While transition rates of women out of employment is large relative to men, these transitions over short period of time (one quarter) may be explained by exit from job market due to motherhood. To understand whether Indian women's weak attachment to the labor market is persistent in future quarters, we compute two quarter transition rates for men and women who were employed in period t but exit employment to non-participation (N) or unemployment (U) in period $t+1$. The gender gap in labor market flows remain persistently high even after two quarters. While 9.7 percent of employed men who transitioned out of employment (to N or U) in previous quarter remain in N for an additional quarter, the majority of women (over 47 percent) remain in N. Women are also behind men in securing employment after one quarter with over 52 percent of employed men in period t transition from N or U in $t+1$ to employment in $t+2$. Only 34 percent of women could achieve a transition back into employment after two quarters.

PLFS data also provides information on whether one or multiple of these three types of job protection are available to employed individuals: a written contract, paid leave, and employer-paid social security. The last part of Panel A presents the transition rates of employed individuals to non-employment (N or U) conditioned on the level of protection available in their job. Access to protective policies, or lack thereof, is an important component of gender gap in employment exits in India. Among employed individuals whose jobs offer at least one of the three protection policies, the gender gap in transition from unemployment to non-participation reduces to 1.9 p.p. compared to the total gap of 4.5 p.p. This gender gap increases significantly to 6 p.p. for jobs with no protection. The quarterly transition rate from employment to unemployment displays a similar pattern. From individuals exiting unemployment from jobs with protections the gender gap is -0.2, while the gap for jobs with no protection is -1.27 p.p. compared to the benchmark gap of -0.8 p.p. Section 5 discussed the role of protection in explaining female labor force participation in India in more details.

Panel B of Table 1 discusses the determinants of the significant gender gap in exit of female labor force in India from employment to U or N. Results are from a linear probability model with dependent variable equals to one if the respondent exits from employment $E \rightarrow \{U, N\}$, and 0 otherwise. We estimate the following linear probability

[2]To contextualize the flow rates in India, we compute the transition rates for men and women using the Census Population Survey Data for the U.S. Within one quarter, 4.6 percent women in the U.S. exit employment to non-participation compared to 2.9 percent men resulting in a gender gap of 1.8 percentage points, i.e., one fourth of the gap observed in India. The gender gap in entry margin is also significantly less striking in the U.S. where transition of women from non-participation to employment is only 2.4 p.p. lower than men, and the transition of women from unemployment to employment is 4.3 p.p. lower than men. In comparison, the gender gap in these rates in India stand at 1.4 p.p. and 9.2 p.p., respectively.

model-

$$\mathbf{1}\{\text{E-N/U}_{i,t+1}\} = \alpha_g + \sum_{k=1}^{3} \beta_{kg} \mathbf{1}\{P_{it} = k\} + X'_{it}\gamma_g + \delta_{s(i),g} + \lambda_{t,g} + \varepsilon_{i,t+1} \tag{1}$$

The dependent variable takes value one for nonparticipation and unemployment next quarter, and zero for all other observed destinations. Due to the large heterogeneity in the types of self-employment available in India, the origin sample includes only regular and casual wage employees, while transition from wage employment to self-employment and unpaid family workers are treated as non-employment destinations and assigned a value of zero.[3] All columns include state and quarter fixed effects, and demographic controls which are gender, age groups (18–24, 25–34, 35–44, 45–54, 55–59), education (up to primary, middle, secondary/higher secondary, and post-secondary diploma or degree), marital-status indicators, and origin household size. Columns 2 and 3 additionally include controls for two-digit industry code, two-digit occupation code, and a dummy indicator of whether the household was in casual employment. Standard errors are clustered by region (district).

Column 1 confirms that there is a large and significant gender gap in employment is exit with the probability of exit for women being nearly 50 percentage points higher than the probability for men. Including controls and interaction of gender dummy with for marital status and education reduces but does not eliminate the gap. Marriage increases the probability of exit for women by 21 p.p. but the gender gap even among unmarried women and men remains high at 34 p.p. Similarly, receiving graduate degree or more reduces the probability of exit for women by 13 p.p. but the gender gap in employment exits remain significant even at the highest levels of education. The transition flows provide a broad picture of movement across labor market states in India and are assessed in more detail in Section 5. However, first we examine the out-of-labor-force destinations associated with non-participation and employment exits by gender and over the life cycle. As women are more likely to leave employment for nonparticipation and less likely to return to work in the following quarter, we next examine the activities they report after leaving employment.

[3]Nearly 1.16% of employed men transition from formal employment to self-employment in a quarter, relative to only 0.58% of employed women. These proportions show that a very small share of E to U or N transitions for women are driven by transitions to self-employment, while the transition to non-employment for men reduces further when accounted for self-employment.

Table 1: Gender Gap in Labor Market Transitions

Panel A. Quarterly transitions and paths, by gender (percent)

	(1) Men	(2) Women	(3) Women – men
Quarterly movements into and out of wage employment			
$E \to N$	0.73	5.21	4.48***
$E \to U$	2.76	1.96	−0.80***
$N \to E$	2.13	0.78	−1.35***
$U \to E$	17.09	7.89	−9.20***
Three-quarter paths after a wage-employment exit			
$E \to N \to N$	9.66	47.41	37.75***
$E \to N \to E$	9.47	22.98	13.51***
$E \to N \to U$	1.56	1.76	0.21
$E \to U \to E$	42.68	11.56	−31.11***
$E \to U \to U$	32.61	11.13	−21.48***
$E \to U \to N$	1.72	3.98	2.26***
Quarterly exits by protections at the origin job			
No recorded protection			
$E \to N$	0.83	6.79	5.97***
$E \to U$	4.15	2.88	−1.27***
At least one recorded protection			
$E \to N$	0.47	2.35	1.88***
$E \to U$	1.04	0.87	−0.16

Panel B. Nonparticipation rather than unemployment following an exit

	Demographics	Full controls	Both interactions
Female	0.513***	0.459***	0.351***
	(0.015)	(0.015)	(0.021)
Married	0.031***	0.036***	−0.017
	(0.010)	(0.010)	(0.011)
Education: middle	0.004	−0.018*	−0.013
	(0.011)	(0.011)	(0.011)
Education: secondary/higher sec.	0.060***	0.012	0.033**
	(0.012)	(0.012)	(0.014)
Education: postsecondary diploma/degree	0.075***	−0.009	0.046**
	(0.015)	(0.014)	(0.018)
Female × Married			0.215***
			(0.018)
Female × Edu: middle			0.014
			(0.022)
Female × Edu: secondary/higher sec.			−0.053**
			(0.025)
Female × Edu: postsecondary diploma/degree			−0.142***
			(0.025)
Regular/casual, two-digit industry and occupation	No	Yes	Yes
Observations	25,087	25,087	25,087
R^2	0.319	0.348	0.360

Notes: Urban PLFS, ages 18–59 at the origin quarter, 2017Q3–2024Q2. Self-employment and unpaid family work are excluded from E. U is unemployment and N is nonparticipation. All columns are weighted by survey weights. Panel B is a linear probability model among $E \to \{U, N\}$ exits. Standard errors in parentheses are clustered at the district level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

4 Exit from Employment: Where do Women Go?

The extensive questionnaire of PLFS allows us to observe the principal status of individuals irrespective of their employment status, and the current status of individuals who were previously employed. We combine these statuses into eight broad categories: Employed (wage/salaried), self-employed, unemployed, childcare and home-making, education, disability, financially well-off, and others.[4]

Panel A and C of Figure 2 shows the usual status of prime-aged women and men, respectively, over their life-cycle. While salaried or self-employment dominates the prime-age men's usual status, women primarily go into home-production i.e. engaging in domestic duties such as home-making and child care. By the age of 27, nearly 73 percent of men are in salaried employment or self-employment while only 20 percent of women between ages 23 and 27 join the workforce. Additionally, over 14 percent of men report being unemployed between ages 22 to 27, compared to only 8 percent of women.[5]

A natural explanation of this gap is a potential gender gap in educational qualifications. If women are significantly less likely to receive education than men, then they're also less capable of participating in the workforce based on lower skill acquisition. However, we observe complete parity in educational pursuits of men and women under the age of 27. By 18, nearly 72 percent of women report their status as currently pursuing education relative to 68 percent of men. On average, between the ages of 18 to 22, women and men are equally likely at approximately 52 percent to report education as their principal status. While these educational qualifications for majority of the men translate into employment, most women transition directly into home-production.

[4]Survey respondents can choose a response out of twelve options which include regular employment, two categories for self-employment, employment as unpaid family worker, casual wage labor, unemployment, education, domestic duties only, domestic duties with free collection of goods like sewing, rentiers/pensioners/remittance recipients, inability to work due to disability, and others (begging, prostitution, etc). We combine the categories of regular and casual employment into Employed, the three categories of self-employment and unpaid labor into Self-employed, and the two category of domestic duties into Childcare/home-making category.

[5]Figure A2 plots the usual and current status of men and women over per-capita consumption. As PLFS only covers urban, middle-income consumers, the range of per-capita income can be classified as lower-middle income to middle income. The patterns over consumption groups is consistent with what we find over life-cycle. Irrespective of past employment status, majority of women across all consumption groups participate in home-production while majority of men are employed. There is only a slight uptick among women's participation in labor market at the highest consumption bracket.

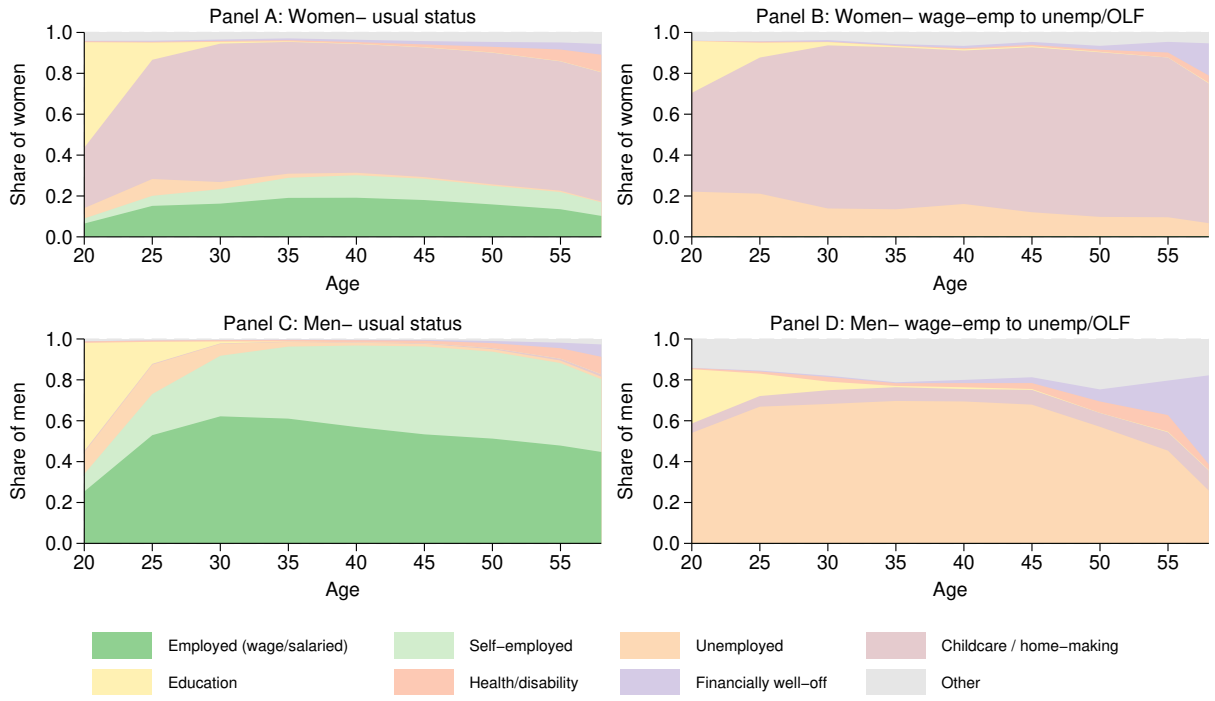


Figure 2: OLF Destinations by Gender and Age

Notes: Figure shows the usual principal status (Panel A and C) and current status of previously employed individuals (Panels B and D) by gender and age. Age is grouped into five-year brackets starting at 18 and ending at 59, and the graph shows the fraction of individuals in a status at each bracket's midpoint.

Another explanation of the gender gap in employment is that women face significant societal and personal constraints regarding participation in workforce even if this attitude is not reflected towards education of women. While we cannot capture societal constraints directly in the data, we can draw some understanding from the current status of women who have previously been employed and are likely to be less constrained by societal expectations than women who have never been employed. Panel B and D of Figure 2 directly compares the current principal status of men and women who were previously employed in wage/salaried jobs. After losing or leaving employment, a significant proportion of men enter unemployment (Panel D) while majority of women enter home-production. 67 percent of young women (between the ages of 23 and 27) enter home-production while only 21 percent enter unemployment and 7.2% pursue education. This patter is reversed among young men who separate from employment, 67 percent of whom report being unemployed, 11 percent pursue education, while only 5% enter into home-production.

These results are consistent with labor force participation patterns in other south-Asian countries. Specifically, Bandiera et al. (2025) show that college-aged men and women in Pakistan are equally ambitious in pursuing employment. However, conditional on their

education and career plans before marriage, women who enter the labor market within two months of graduation are more likely to continue employment while women who postpone labor market participation until marriage face larger societal restrictions in pursuing future employment.

While marital status can explain a substantial share of women's labor supply preferences, but it is not the sole determinant of women's decisions to enter or exit the workforce. Appendix Figure A1 plots the current status of married women (Panel A) and unmarried women (Panel B) who separate from employment. Between the ages of 23 and 27, 84 percent of married women leave employment and enter home-production while only 11 percent enter unemployment and less than 1 percent pursue education. Among unmarried women, transitions into home production and unemployment are equally common, each accounting for 37 percent of employment separations. Nearly 18 percent of unmarried women transition into education following separation from employment. However, the share of unmarried women in home-production increases with age. We formally test the role of marriage in explaining women's transition out of labor force in Table 1 and find that married women are 21 percent more likely to exit the labor force relative to unmarried women. However, even unmarried women are significantly more likely at 34 percent to exit out of labor force relative to men.

Our findings suggest that, irrespective of marital status, women are disproportionately more likely to be responsible for household duties in India. This is puzzling as marriage and child-bearing has been identified as one of the main reasons for lower labor force participation rate of women, especially in emerging countries, and for the higher participation of women in informal sectors (Bandiera et al., 2025; Berniell et al., 2021). To better understand the reason for this disproportionate participation in home-production for all women in India, we turn towards causally assessing the effect of quality of jobs available on male and female labor force participation.

5 Role of Employment Protection

In this section we study whether the weak labor force attachment of women is driven by the quality of employment available. The answer is yes. As documented in Table 1, a significantly larger fraction of women transition from employment to out of labor force if they are employed in jobs which offer no protections, relative to jobs that offer multiple protections. Table 2 tests this idea formally by estimating the effect of employment protections on the probability of transitioning from employment to non-participation for men and women using a weighted logit model.

As our focus is to study the role of protection on employment attachment, the dependent variable in Table 2 takes value one for transition from employment to non-participation next quarter and zero for all other observed destinations, including unemployment and self-employment. We define protection count P_{it} as the number of protections available in the job for employed individuals. Protections can include one of these three options: written contract, paid leave, and employer social security. All columns control for individual level demographic characteristics in X_{it} including age group, education, and marital status, as well as state and quarter fixed effects. Columns 2 and 3 additionally include controls for two-digit industry and two-digit occupation codes.

Table 2: Job protections and quarterly employment exits: Odds ratios

	(1) Demographics		(2) Full controls	
	Men	Women	Men	Women
A. Odds ratios relative to zero protections				
1 of 3 protections	0.713**	0.546***	0.853	0.603***
	(0.114)	(0.066)	(0.142)	(0.076)
2 of 3 protections	0.747**	0.362***	0.925	0.402***
	(0.105)	(0.064)	(0.136)	(0.073)
3 of 3 protections	0.628***	0.279***	0.815	0.322***
	(0.081)	(0.037)	(0.120)	(0.047)
B. Change in exit odds (%)				
2 versus 1 protection	4.6	−33.6**	8.4	−33.3**
	(17.7)	(13.0)	(19.1)	(13.4)
3 versus 1 protection	−12.0	−48.9***	−4.5	−46.6***
	(15.2)	(7.8)	(17.8)	(8.6)
Demographics and household size	Yes	Yes	Yes	Yes
Regular/casual, industry and occupation	No	No	Yes	Yes
Observations	160754	46683	160142	46593

Notes: Results from a weighted logit model with dependent variable coded as 1 for individuals exiting from employment to non-participation, by gender. Protections are a written contract, paid leave and employer-provided social security. All regressions use origin-interview survey weights. All columns include demographic controls, state and origin-quarter fixed effects, while Columns 2 and 3 additionally control for regular/casual employment and two-digit industry and occupation indicators from the origin interview's current-weekly job. Standard errors are clustered by district. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The main takeaway is that employment protections matter significantly more for labor market participation of women than for men and this effect is monotonic with higher

protections resulting in significantly lower non-participation by women. The probability of moving from employment to non-participation decreases by 33 percent for women if their jobs include one additional protection, and nearly 50 percent if their jobs include two additional amenities and protections. In stark contrast, increase in employment protections has no significant effect on the probability of non-participation for men.[6] These results continue to hold when controlling for industry and occupation fixed effect (Columns 3 and 4).[7]

Additional descriptive evidence presented in Table A2 suggests that unemployed women incur significant job searching efforts with 55 percent women searching through direct application to job ads relative to 46 percent of men. Similarly, 13 percent of women and 8 percent of men search by registering at the employment exchange. In contrast, more men seek financial and job search help from personal networks. However, despite a higher direct search effort, women spend longer under unemployment spell than men, potentially holding out for higher quality jobs. This distinction is also highlighted in the principal location of workplace reported by women. The majority of employed women work at jobs which are conducted at either the employer's enterprise, employer's dwellings, or at their own dwelling unit. These may include formal jobs at companies or informal jobs at households like domestic help, nanny, or cook. Working at established structures likely provide better facilities such as safe work environment and access to rest areas compared to jobs that require traveling. As a result, men are twice as likely as women to work in jobs that lack a structured workplace.

Given the preference of women for better quality jobs, the weaker labor market dynamics of female labor force in India relative to the female workforce in other developing countries may then be explained by the lack of availability for high quality job opportunities. Indeed India performs very poorly in job quality indicators relative to other developing countries. According to the Global Job Quality data constructed by the World Bank (Hovhannisyan et al., 2022), India ranks 39 for job quality in a panel of 40 developing

[6]Table A1 presents the result from two weighted logit models with dependent variable as $\mathbf{1}\{\text{E-N}_{i,t+1}\}$ in Panels A and B, and as $\mathbf{1}\{\text{E-U}_{i,t+1}\}$ in Panels C and D. The table excludes competing exits, i.e., all individuals who transitioned from employment to unemployment are excluded from Panels A and B, and all individuals who exited out of labor force are excluded from Panel C and D. The results show that while the sensitivity of labor force participation to protections are unique to women (result in Panels A and B are similar to Table 2), protections also matter significantly for transition from employment to unemployment for both men and women. The probability of transition to unemployment when offered two additional protections is 60 percent lower for men and 44 percent lower for women.

[7]The jobs offering the greatest employment protections in India are typically in the public sector, where transitions from employment to non-participation are significantly less common for both men and women. Although controlling for industry and occupation fixed effects helps address concerns that our results are driven by public sector employment, we further re-estimate the model after excluding all public sector workers from the sample. The results remain unchanged.

countries, significantly behind countries like Brazil, Vietnam, Indonesia, and Philippines. We now use the estimated relationship between job protection and labor market participation to assess how labor force participation would evolve under alternative counterfactual scenarios, including different levels of employment protection.

6 Implications for Participation

So far, we have documented that women are more likely than men to leave employment and enter nonparticipation, and that these differences are smaller among workers with employment protections. We now assess how female labor force participation would change if women faced the transition rates observed among protected workers or among men.

Table 3 reports the long-run participation and employment rates implied by these alternative flows.[8] In each counterfactual, we replace selected female transition probabilities and adjust the probability of remaining in the origin state so that probabilities sum to one. All other transitions remain fixed. Under women's own transition rates, long-run participation is 19 percent and wage employment is 11 percent, and exercise starts from this benchmark.

Panel A examines the role of employment protection for the exit margin. The exercise points to the importance of employment protections and in the Indian context, even just formalizing the labor market. If unprotected women remained in employment at the rates observed among protected women, female participation would increase to 23 percent and wage employment to 16 percent, increases of 21 and 40 percent relative to the benchmark. These gains are large and arise entirely from improved retention, with no change along the entry margin. Matching protected women's exit rates to those of protected men yields smaller gains, with participation reaching 21 percent and employment 13 percent, whereas extending men's overall exit rates to all female employees raises participation to 31 percent and employment to 21 percent. Overall this suggests that policies that offer more employment protection can help women to stay in the labor force and thus improve overall participation.

Panels B and C point to the importance of both the entry margin for women as well as their retention in the labor force. If women entered employment from nonparticipation at the rate that men do, female participation would increase to 30 percent and wage

[8]We distinguish wage/casual employment (E), self-employment including unpaid family work (S), unemployment (U), and nonparticipation (N). Labor force participation includes E, S, and U, while the employment column reports wage/casual employment (E) only.

employment to 21 percent. Matching women's employment-to-nonparticipation rate to men's produces even larger gains, with participation reaching 33 percent and employment 24 percent. These gains suggest considerable scope for increasing participation through both margins. Improving retention is particularly relevant in the Indian context, where, as we document earlier, women who exit employment often remain outside the labor force subsequently. Policies that help women sustain employment could therefore reduce the repeated loss of workers who have already entered the labor market.

Table 3: Counterfactual Exercises: Implications for Participation

	LFPR	Employment
Benchmark	19.3	11.1
Panel A. Protection: Exit Margin		
Protected women	23.4	15.5
Protected men	20.8	12.5
All men	30.9	21.3
Panel B. Flow Rates: Entry into Employment		
U to E	20.1	12.6
N to E	30.3	21.4
Panel C. Flow Rates: Exit from Employment		
E to U	18.9	10.5
E to N	33.3	24.2

Notes: Entries in this table are implied steady-state rates for women aged 18–59, in percent. Panel A replaces employment exit rates with those of protected women, protected men, or all men, as indicated. Protected jobs offer at least one of three benefits: a written contract, paid leave, or social security. The first two exercises apply to unprotected and protected women, respectively. Panels B and C replace the indicated transition rate with the corresponding male rate. All other transitions remain fixed, except the probability of staying in the same state.

The gains from changing unemployment transitions are smaller. Matching women's unemployment-to-employment rate to men's raises participation to 20 percent and wage employment to 13 percent, whereas matching the employment-to-unemployment rate lowers them to 19 and 11 percent. Helping unemployed women find jobs improves employment, but these women already belong to the labor force. Its effect on participation depends on whether finding work subsequently reduces their likelihood of withdrawing.

The larger gains from movements between employment and nonparticipation suggest that increasing female participation requires attention to women who leave the labor force or remain outside it, alongside those actively seeking work.

Our findings suggest a role for expanding formal employment in increasing female participation. Given that women often report domestic duties after leaving work, access to paid leave and greater job security may help them manage household responsibilities while remaining employed. This is consistent with evidence that maternity leave supports employment continuity after childbirth (Waldfogel, 1998), and that more generous paid family leave increases the likelihood of returning to the previous employer among women who return to employment (Bana et al., 2020). In India, Chakraborty and Mahajan (2026) document higher female employment shares in larger firms and point to amenities such as maternity benefits, transport, and paid leave as possible explanations. Our counterfactual indicates the quantitative scope for improved retention: matching eligible unprotected women's exit rates to those of protected women raises implied participation by 21 percent, even with no improvement in entry. These results suggest that helping women sustain employment could contribute substantially to higher participation.

7 Conclusion

We use rich micro-data to study how employment status transitions can account for India's puzzling low female labor force participation rate. While a substantial literature explores the role of barriers to entry into paid employment, we establish the importance of the exit margin. Indian women transition out of paid employment at higher rates than men, even conditional on a rich set of covariates including marriage and age. We show that upon transitioning to non-employment, Indian women engage primarily in home production and have much lower rates of re-entry into the labor force in subsequent quarters.

We find that for women, the presence of employment protections, measured as formal contracts, paid leave or social security benefits, can substantially reduce the likelihood of an exit to non-employment. A simple back of the envelope calculation shows that replacing female transition rates by those of women with at least one employment protection could raise female labor force participation by nearly a fifth of its baseline level.

Our paper establishes an urgent need for further work studying the link between employment protection and labor market outcomes for different gender groups. Important extensions of our work include evaluations of specific forms of employment protection for women and a further analysis of the mechanisms through which these protections work. For instance, Ho et al. (2026) show that allowing working from home can dramati-

cally raise labor force participation, and emphasize the entry margin and possibility of transitions to formal employment. It would be interesting to see whether such amenities also translate to higher retention on the job. Our findings suggest that understanding the policy responses that can accelerate formalization and the rapid diffusion of protections through an economy undergoing rapid growth and transformation merit more analysis. Finally, we show that understanding the role of the exit margin is a fruitful avenue for explaining cross-country differences in participation more broadly, and plan to pursue this in ongoing research.

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A Appendix: Additional Figures and Tables

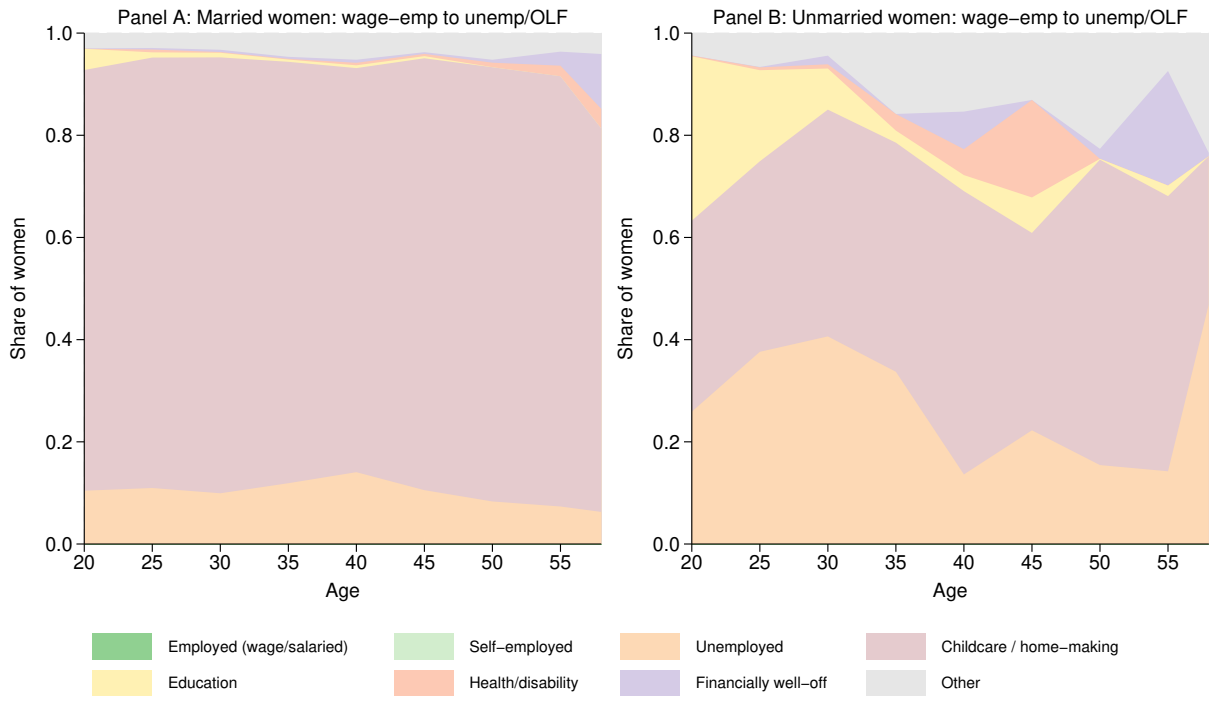


Figure A1: Reason for Leaving Employment - By Marital Status

Notes: Figure shows the current status of previously employed women by marital status and age. Age is grouped into five-year brackets starting at 18 and ending at 59, and the graph shows the fraction of individuals in a status at each bracket's midpoint.

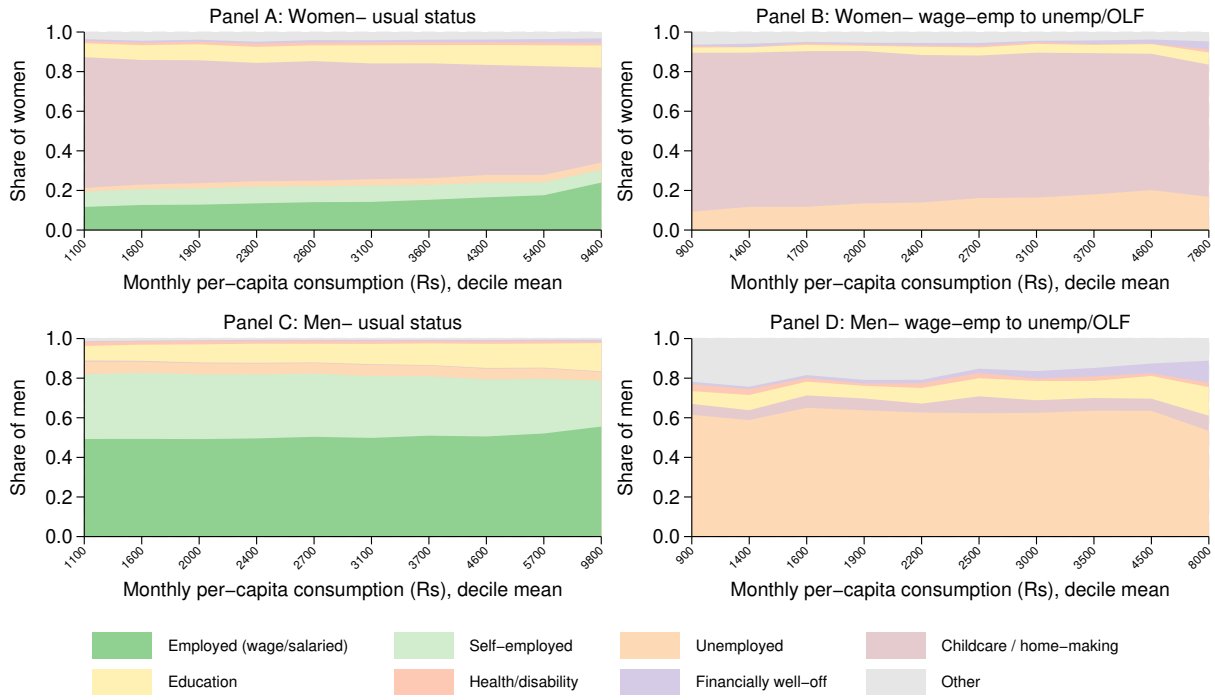


Figure A2: OLF Destinations by Per-Capita Consumption

Notes: Figure shows the usual principal status (Panel A and C) and current status of previously employed individuals (Panels B and D) of women and men, respectively, at different levels of consumption.

Table A1: Job protections and quarterly employment exits: Competing exits excluded

	(1) Demographics		(2) Full controls	
	Men	Women	Men	Women
A. $E \to N$: Odds ratios relative to zero protections				
1 of 3 protections	0.681**	0.529***	0.848	0.589***
	(0.110)	(0.065)	(0.142)	(0.075)
2 of 3 protections	0.706**	0.349***	0.917	0.391***
	(0.100)	(0.062)	(0.135)	(0.071)
3 of 3 protections	0.592***	0.271***	0.804	0.316***
	(0.078)	(0.036)	(0.120)	(0.046)
B. $E \to N$: Change in exit odds (%)				
2 versus 1 protection	3.6	−34.1**	8.1	−33.7**
	(17.6)	(13.0)	(19.1)	(13.4)
3 versus 1 protection	−13.2	−48.8***	−5.2	−46.4***
	(15.1)	(7.9)	(17.8)	(8.8)
C. $E \to U$: Odds ratios relative to zero protections				
1 of 3 protections	0.484***	0.470***	0.951	0.732*
	(0.050)	(0.086)	(0.094)	(0.135)
2 of 3 protections	0.324***	0.367***	0.743**	0.518***
	(0.046)	(0.079)	(0.108)	(0.112)
3 of 3 protections	0.193***	0.262***	0.467***	0.392***
	(0.036)	(0.083)	(0.091)	(0.131)
D. $E \to U$: Change in exit odds (%)				
2 versus 1 protection	−33.1***	−21.9	−21.9	−29.2
	(10.3)	(18.5)	(13.0)	(17.1)
3 versus 1 protection	−60.1***	−44.2**	−50.9***	−46.4**
	(8.0)	(14.6)	(10.0)	(14.2)
Demographics and household size	Yes	Yes	Yes	Yes
Regular/casual, industry and occupation	No	No	Yes	Yes
State and quarter FEs	Yes	Yes	Yes	Yes
Clustering	District	District	District	District
Observations: $E \to N$	156927	45972	156327	45883
Observations: $E \to U$	157788	44319	157364	44225

Notes: Separate weighted logits for exits from wage/casual employment (E) to nonparticipation (N, Panels A–B) and unemployment (U, Panels C–D). Panels A–B exclude unemployment destinations; Panels C–D exclude nonparticipation destinations. Each outcome is one for its named destination and zero for continued wage/casual employment or self-employment (including unpaid family work). These zero-outcome observations are common before model-specific perfect-prediction exclusions. Eligible workers are urban employees aged 18–59, visits 1–2, with complete protection responses and aligned usual/current job types. Protections are a written contract, paid leave and social security. Each model interacts female with protection count; other control slopes are shared. Origin survey weights are used. Parentheses report district-within-state clustered standard errors on the displayed scale. Panels A and C report odds ratios relative to zero protections. Panels B and D report $100[\exp(\beta_k - \beta_1) - 1]$: percentage changes in odds relative to one protection, not changes in exit probabilities. Standard errors use the delta method. Perfect prediction excludes 28/717 observations for $E \to N$ and 2454/2972 for $E \to U$ in specifications (1)/(2). Associations are not causal effects. Stars test zero log-odds contrasts: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A2: Employment Search and Workplace Location (weighted %)

	Men	Women
Panel A: Duration of unemployment spell		
≤ 6 months	13.7	11.9
>6 months – 1 year	29.4	26.7
>1 – 2 years	29.1	25.5
>2 – 3 years	14.1	15.7
>3 years	13.8	20.3
Observations (N)	40,963	22,691
Panel B: Efforts undertaken to search work		
apply to prospective employers/place or answer job advertisements/check at factories, work sites	45.8	55.4
register with employment exchange	7.8	12.9
register with private employment center	6.2	5.0
seek financial help to start a business	3.3	0.8
seek help from relatives, friends	18.7	12.5
apply for permit or license to start a business	0.5	0.2
others	17.7	13.3
Observations (N)	43,600	23,812
Panel C: Principal location of workplace		
Rural: all	2.5	2.9
Urban: own dwelling unit	3.6	14.2
Urban: attached structure	2.3	4.9
Urban: open area adjacent	0.5	0.9
Urban: detached structure	1.0	1.7
Urban: own enterprise away	17.2	6.9
Urban: employer's dwelling	2.1	13.4
Urban: employer's enterprise	47.4	48.1
Street (fixed location)	3.6	2.0
Construction site	8.6	2.4
Other (urban)	3.3	0.9
No fixed workplace	7.8	1.8
Observations (N)	1,137,883	286,832

Table A3: Type of Employment Benefits (weighted %)

	Men	Women
Panel A: Type of job contract		
No written contract	70.6	67.2
Written ≤ 1 year	5.5	5.8
Written 1–3 years	3.8	4.6
Written >3 years	20.2	22.4
Observations (N)	683,674	188,158
Panel B: Enterprise size (no. of workers)		
<6 workers	54.5	55.1
6–9 workers	11.5	9.5
10–19 workers	7.2	8.3
>= 20 workers	22.9	24.2
Not known	4.0	2.9
Observations (N)	1,157,806	295,029
Panel C: Social security benefits		
PF/pension only	8.4	8.4
Gratuity only	0.4	0.3
Health care/maternity only	0.6	0.9
PF + Gratuity	3.5	3.3
PF + Health care	6.6	7.4
Gratuity + Health care	1.2	1.2
PF + Gratuity + Health care	17.7	19.0
Not eligible	57.5	56.5
Not known	4.2	3.0
Observations (N)	682,863	188,038

B Online Appendix

Table B4: Quarterly transitions between wage employment and own-account or employer work

	Men	Women	Women – men
$E \to S_1$	1.16	0.58	−0.58***
			(0.06)
Origin observations	483,374	140,616	
$S_1 \to E$	2.20	1.61	−0.60***
			(0.11)
Origin observations	279,700	60,336	

Notes: Rates are percentages of observed quarterly transitions from the indicated origin state, calculated separately by gender using origin-interview survey weights. The gap and its standard error (in parentheses) are in percentage points. The sample matches Table 1's quarterly transition sample: urban PLFS respondents aged 18–59 at the origin interview, observed in consecutive quarters, with destination interviews from 2017Q4 to 2024Q2. E is regular/casual wage employment (CWS 31, 41, 42, 51, 71, 72 and 98). S_1 consists of own-account workers and employers (CWS 11 and 12); unpaid family workers (21) and household-enterprise temporary-absence codes (61 and 62) are not classified as S_1. The denominator for $E \to S_1$ includes all observed E-origin transitions; the denominator for $S_1 \to E$ includes all observed S1-origin transitions. Other destinations remain in these denominators. Observation counts are unweighted person-quarter transitions, not unique workers. Gender gaps are estimated without controls, with standard errors clustered by district within state. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table B5: Quarterly U.S. labor-market transition rates

Transition	Full sample	Men	Women	Women – men
$E \to N$	3.73	2.86	4.66	+1.80
$E \to U$	1.94	2.00	1.88	-0.12
$N \to E$	8.33	9.83	7.41	-2.42
$U \to E$	34.49	36.47	32.15	-4.31

Notes: Entries are percentages. The source is the harmonized U.S. Current Population Survey series assembled by Donovan, Lu, and Schoellman. The sample covers urban respondents ages 16–65 and reports quarterly transitions averaged over 2010–2021. E denotes wage or salaried employment, U unemployment, and N nonparticipation. Self-employment is a separate, unreported state and is not included in E. Full-sample rates pool men and women within each year using their origin-state population shares before averaging across years.

Table B6: Job protections and quarterly exits: excluding public employment

	(1) Demographic controls		(2) Full controls	
	Men	Women	Men	Women
A. Relative to zero protections				
1 of 3 protections	−1.983***	−4.180***	0.124	−3.241***
	(0.273)	(0.650)	(0.239)	(0.637)
2 of 3 protections	−2.254***	−5.066***	0.011	−4.588***
	(0.318)	(0.687)	(0.281)	(0.668)
3 of 3 protections	−2.204***	−5.509***	0.137	−4.994***
	(0.298)	(0.536)	(0.260)	(0.514)
B. Differences across protection counts				
2 protections − 1 protection	−0.271	−0.886	−0.113	−1.348*
	(0.265)	(0.757)	(0.283)	(0.738)
3 protections − 1 protection	−0.220	−1.329**	0.013	−1.753***
	(0.270)	(0.618)	(0.259)	(0.630)
Regular/casual employment	No	No	Yes	Yes
Two-digit industry and occupation FEs	No	No	Yes	Yes
Observations	121,025	31,490	121,025	31,490

Notes: Results from a linear probability model with dependent variable coded as 1 for individuals exiting from employment to non-participation or unemployment, by gender. Protections are a written contract, paid leave and employer-provided social security. All regressions use origin-interview survey weights. All columns include demographic controls, state and origin-quarter fixed effects, while Columns 2 and 3 additionally control for regular/casual employment and two-digit industry and occupation indicators from the origin interview's current-weekly job. Standard errors are clustered by district. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Public employment is identified at visit 1: enterprise codes 5 (government/local bodies), 6 (public-sector enterprises) and 7 (autonomous bodies) are excluded. Public/private limited companies (8), other enterprises and missing ownership remain.

Table B7: Counterfactual Table with Self-Employment

	LFPR	Employment
Benchmark	19.3	10.8
Panel A. Protection and the Exit Margin		
Protected women	23.8	16.0
Protected men	20.6	11.9
All men	28.3	17.8
Panel B. Flow Rates: The Entry Margin		
S+U to E	20.0	12.4
N to E	30.3	20.9
Panel C. Flow Rates: The Exit Margin		
E to S+U	18.6	9.5
E to N	32.5	22.9

Notes: Values are implied steady-state levels among Indian women, in percent. The sample comprises urban PLFS respondents aged 18–59 at the origin interview. States are wage/casual employment (E), self-employment and unemployment pooled (S+U), and nonparticipation (N). Self-employment includes unpaid family work. LFPR includes E and S+U; Employment denotes wage/casual employment only.